

Problem Solving in Geometry (II)

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Some Examples on Circles and Tangents

1. Let BC be a chord of a circle Γ , and let S_1, S_2 be the two arcs of Γ cut off by BC . Let M be the mid-point of S_2 . A circle Ω touch S_1 and BC . Show that the tangent from M to Ω equals MB .

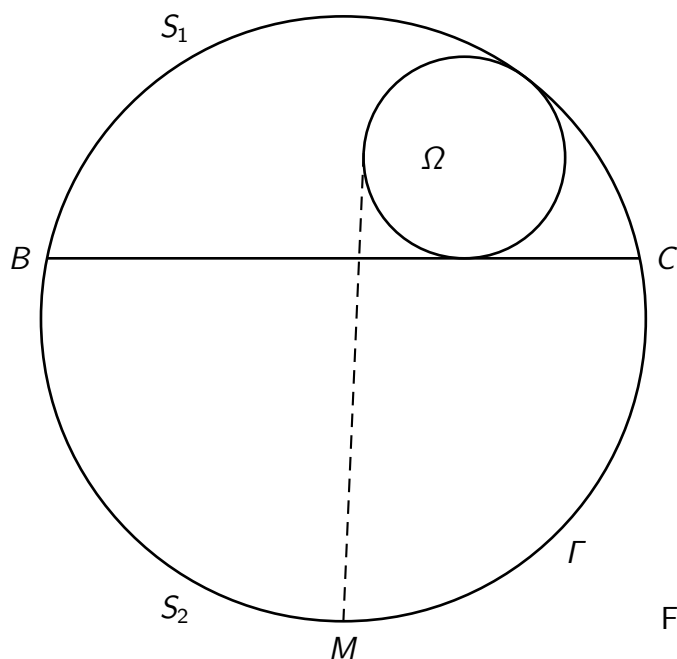


Figure 1

2. Let ABC be a triangle with circumcircle Γ . Let D be a point on BC . A circle Ω touches Γ , DC and AD . Let E and F be the point of contact of Ω on DC and AD . Show that EF passes through the in-centre of $\triangle ABC$.

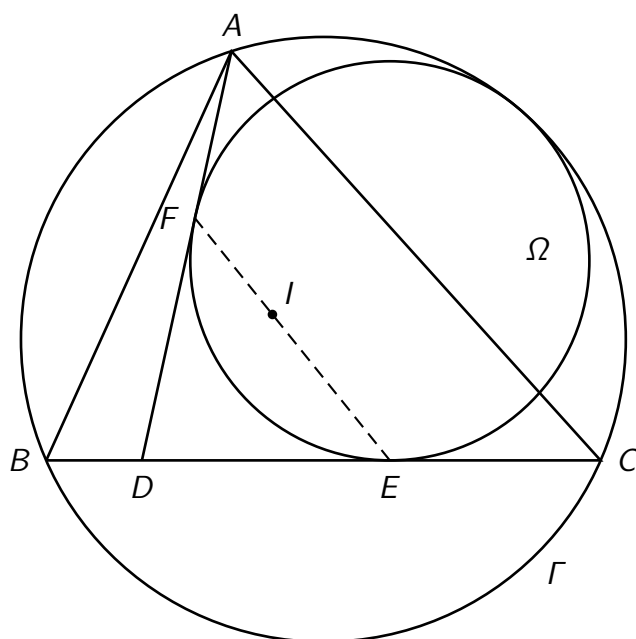


Figure 2

3. Thébault's Theorem

Let ABC be a triangle with circumcircle Γ . Let D be a point on BC . Let Ω_1 be the circle touches Γ , BD and AD . Let Ω_2 be the circle that touches Γ , DC and AD . Let P and Q be the centres of Ω_1 and Ω_2 respectively. Show that P , Q and I are colinear, where I is the in-centre of $\triangle ABC$.

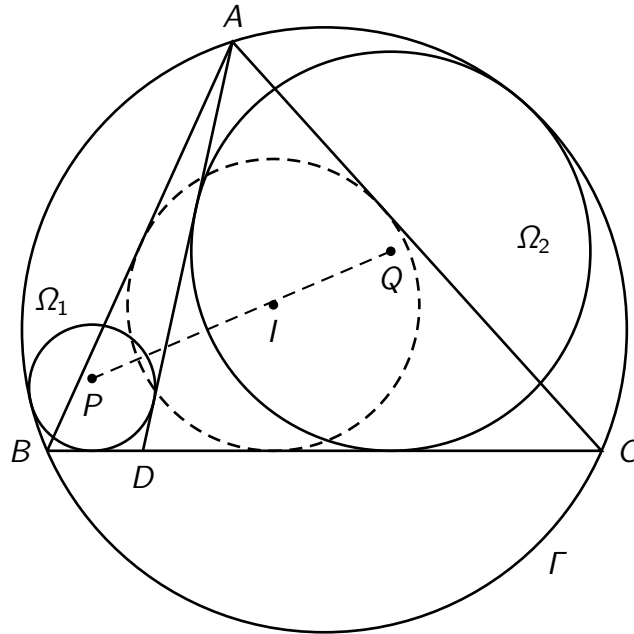


Figure 3

4. Casey's Theorem – The Generalized Ptolemy Theorem

Let circles $\Omega_1, \Omega_2, \Omega_3, \Omega_4$ (in cyclical order) all touch the circle Γ , Then

$$\widehat{\Omega_1\Omega_2} \cdot \widehat{\Omega_3\Omega_4} + \widehat{\Omega_2\Omega_3} \cdot \widehat{\Omega_4\Omega_1} = \widehat{\Omega_1\Omega_3} \cdot \widehat{\Omega_2\Omega_4},$$

where $\widehat{\Omega_i\Omega_j}$ denotes

- (a) the common external tangent of circles Ω_i and Ω_j if both circles touch Γ from the same side;
- (b) the common internal tangent of circles Ω_i and Ω_j if Ω_i and Ω_j touch Γ from opposite sides;
- (c) the tangent to Ω_j from the Ω_i if the circle Ω_i is degenerated into a point;
- (d) the length $\Omega_i\Omega_j$ if both circles Ω_i and Ω_j are degenerated into points.

Proof

Let O_i and O_j be the centres of two circles Ω_i and Ω_j with radii r_i and r_j respectively. Then

$$\text{the length of the common **external** tangents} = \sqrt{O_iO_j^2 - (r_i - r_j)^2},$$

$$\text{the length of the common **internal** tangents} = \sqrt{O_iO_j^2 - (r_i + r_j)^2}.$$

Let K_i denotes the point of contact of Ω_i to Γ and let O and r be the centre and radius of Γ respectively. Suppose both Ω_i and Ω_j touches Γ externally. Then by Cosine Formula,

$$K_iK_j^2 = 2r^2 - 2r^2 \cos \angle O_iOO_j$$

But

$$\begin{aligned} O_iO_j^2 &= (r + r_i)^2 + (r + r_j)^2 - 2(r + r_i)(r + r_j) \cos \angle O_iOO_j \\ &= (r + r_i)^2 + (r + r_j)^2 - 2(r + r_i)(r + r_j) \left(1 - \frac{K_iK_j^2}{2r^2}\right) \\ &= [(r + r_i) - (r + r_j)]^2 + \frac{(r + r_i)(r + r_j)}{r^2} K_iK_j^2 \\ &= (r_i - r_j)^2 + \frac{(r + r_i)(r + r_j)}{r^2} K_iK_j^2 \end{aligned}$$

Hence

$$\widehat{\Omega_i\Omega_j} = \frac{K_iK_j}{r} \sqrt{(r + r_i)(r + r_j)}.$$

By similar argument, if both Ω_i and Ω_j touches Γ internally,

$$\widehat{\Omega_i\Omega_j} = \frac{K_iK_j}{r} \sqrt{(r - r_i)(r - r_j)}.$$

And if Ω_i touches Γ internally and Ω_j touches Γ externally,

$$\widehat{\Omega_i\Omega_j} = \frac{K_iK_j}{r} \sqrt{(r + r_i)(r - r_j)}.$$

Hence

$$\begin{aligned}
& \widehat{\Omega_1\Omega_2} \cdot \widehat{\Omega_3\Omega_4} + \widehat{\Omega_2\Omega_3} \cdot \widehat{\Omega_4\Omega_1} \\
&= \frac{K_1K_2 \cdot K_3K_4 + K_2K_3 \cdot K_4K_1}{r^2} \sqrt{(r \pm r_1)(r \pm r_2)(r \pm r_3)(r \pm r_4)} \\
&= \frac{K_1K_3 \cdot K_2K_4}{r^2} \sqrt{(r \pm r_1)(r \pm r_2)(r \pm r_3)(r \pm r_4)} \\
&= \widehat{\Omega_1\Omega_3} \cdot \widehat{\Omega_2\Omega_4}
\end{aligned}$$

Here, the sign of $\pm r_i$ depends on the internal/external tangency of the four circles with Γ and

$$K_1K_2 \cdot K_3K_4 + K_2K_3 \cdot K_4K_1 = K_1K_3 \cdot K_2K_4$$

follows from the Ptolemy's Theorem.

Remarks The converse of the Generalized Ptolemy's Theorem is also true. Namely, if

$$\widehat{\Omega_1\Omega_2} \cdot \widehat{\Omega_3\Omega_4} + \widehat{\Omega_2\Omega_3} \cdot \widehat{\Omega_4\Omega_1} = \widehat{\Omega_1\Omega_3} \cdot \widehat{\Omega_2\Omega_4},$$

then there exists a fifth circle Γ that touches all of them (with the appropriate combination of internal/external tangency). Inversion is used in the proof of the converse theorem.

5. Application of Casey's Theorem

Proof of Problem 1.

Let P and Q be the contact points of Ω to Γ and BC respectively. Applying Casey's Theorem on B, Ω, C and M , we have

$$BQ \cdot CM + CQ \cdot BM = \widehat{M\Omega} \cdot BC.$$

But $BM = CM$ and $BQ + CQ = BC$, it follows that $\widehat{M\Omega} = BM$.

6. Application of the Converse of Casey's Theorem

Feuerbach's Theorem The nine-point circle is tangent the the incircle and the three ex-circle of a triangle.

Let a, b, c be the sides of $\triangle ABC$, D, E, F be the mid-points of AB, BC, CA and Ω be the incircle. Consider the circles D, E, F, Ω , we have

$$\begin{aligned}
\widehat{DE} &= \frac{c}{2}, \quad \widehat{EF} = \frac{a}{2}, \quad \widehat{FD} = \frac{b}{2}, \\
\widehat{D\Omega} &= \left| \frac{b-c}{2} \right|, \quad \widehat{E\Omega} = \left| \frac{c-a}{2} \right|, \quad \widehat{F\Omega} = \left| \frac{a-b}{2} \right|.
\end{aligned}$$

But

$$a(b-c) + b(c-a) + c(a-b) = 0$$

So for some ordering of D, E, F, Ω , the six tangents satisfy the condition. Hence there is a circle touching the incircle Ω and passing through D, E, F and this circle is the nine-point circle. For the excircle Ω_A outside BC , we have

$$\widehat{D\Omega_A} = \left| \frac{b-c}{2} \right|, \quad \widehat{E\Omega} = \left| \frac{a+c}{2} \right|, \quad \widehat{F\Omega} = \left| \frac{a+b}{2} \right|.$$

But

$$b(a+c) - c(a+b) = a(b-c)$$

By the Converse of Casey's Theorem, the nine-point circle is also tangent to the excircle Γ_A . Similarly, the nine-point circle also touches the other two excircles.

7. Cleavers

- (a) Let ABC be a triangle with $AB < AC$. Let Ω be the circumcircle of ABC . Let M be the midpoint of the arc CB of Ω containing A . Let D be the foot of perpendicular from M to AC . Prove that $BA + AD = DC$.

Note that $A'D$ bisects the perimeter of the triangle, where A' is the midpoint of BC . This is call **cleaver** of a triangle.

- (b) Show that the three cleavers are concurrent.

8. Splitters

A splitter AD bisects the perimeter of the triangle.

(a) Show that the three splitters are concurrent.

(b) Let the Nigel point M be point of concurrent of the three splitters, show that I, G, M are collinear.

9. Cool Down

Let I be the incentre of $\triangle ABC$. E and F be the points of contact of the incircle on BC and AC . Let AI and EF intersect at X . Prove that $\angle AXB = 90^\circ$.